

Integrated Circuit ECE326

Lec. 10

CMOS inverter Characteristics

Dr. Ahmed Reda Mohamed^{1,2}

¹Ph.D, the MoE Key Laboratory of Artificial Intelligence & Bio-Circuits and Systems Laboratory (*BiCASL*), Department of Micro-Nano Electronics, Shanghai Jiao Tong University, Shanghai, China.

²Assistant professor, Department of Electronics and Communications, Zagazig University, Zagazig , Egypt.



- ☐ DC transfer Ch/CS Curve
- ☐ Noise Margin Concept
- ☐ MOS Inverter Topologies
- ☐ DC Transfer Characteristics
- ☐ Beta Ratio Effects
- ☐ Dynamic Characteristics of CMOS Inverter
- ☐ Inverter switching characteristics
- ☐ Power Dissipation
- ☐ Discussion



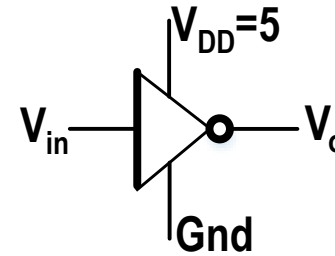
- ☐ **DC transfer Ch/CS Curve**
- ☐ **Noise Margin Concept**
- ☐ **MOS Inverter Topologies**
- ☐ **DC Transfer Characteristics**
- ☐ **Beta Ratio Effects**
- ☐ **Dynamic Characteristics of CMOS Inverter**
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- ☐ **Power Dissipation**
- ☐ **Discussion**

DC transfer Ch/Cs curve



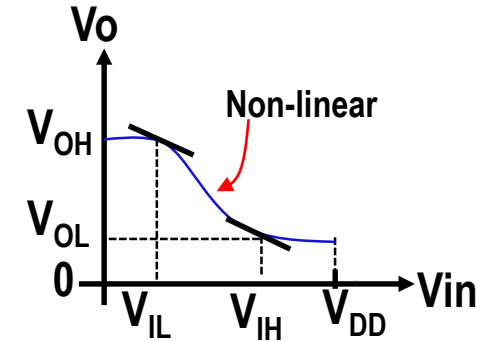
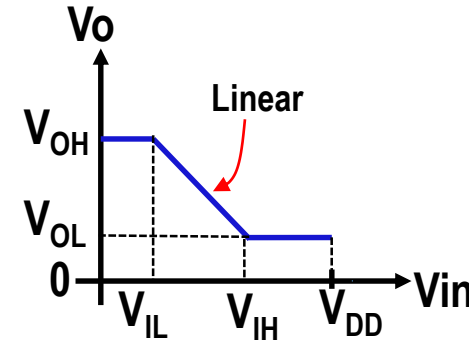
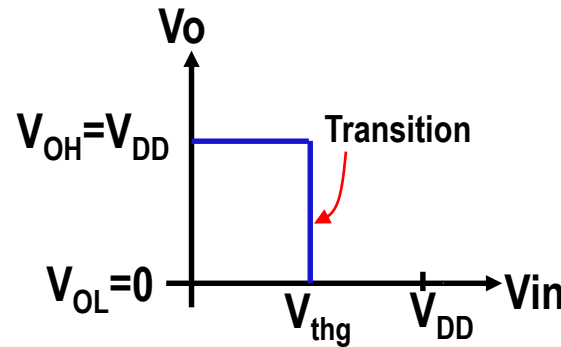
Definition

It is a relation between V_{in} and V_o



Types of Ch/Cs curve

- Ideal curve
- Simplified curve
- Practical curve



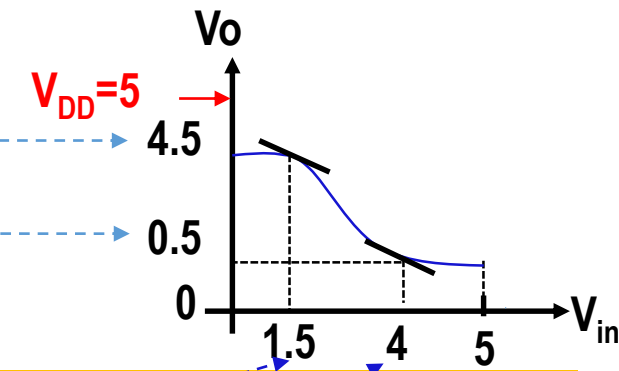
Notes

V_{OH} : Minimum output voltage can be considered as “1”

V_{OL} : Maximum output voltage can be considered as “0”

V_{IL} : Maximum input voltage can be considered as “0”

V_{IH} : Minimum input voltage can be considered as “1”



Ideally, $V_{OH} = V_{DD}$, $V_{OL} = GND$
 $V_{IH} = V_{IL} = V_{DD}/2$



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Noise Margin Concept (1/2)



Definition

It indicates how system can immune against the noise

Categories of Noise Margin

$NMH = V_{OH} - V_{IH}$

$NML = V_{IL} - V_{OL}$

Remember

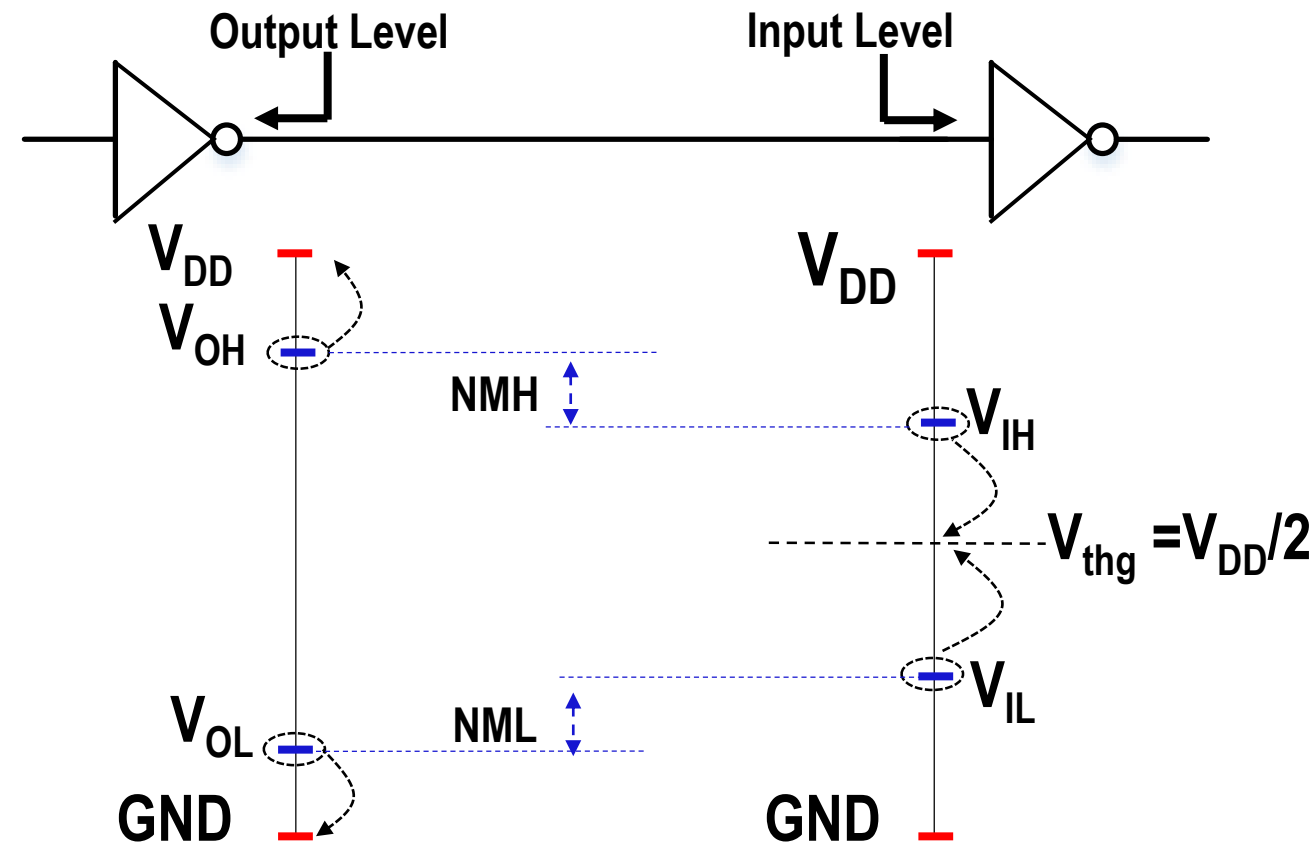
Ideally, $V_{OH} = V_{DD}$, $V_{OL} = GND$

$V_{IH} = V_{IL} = V_{DD}/2$



$NMH = V_{DD}/2$

$NML = V_{DD}/2$



Noise Margin Concept (2/2)



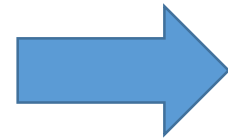
□ Ideal CMOS inverter

□ $NMH = V_{OH} - V_{IH}$

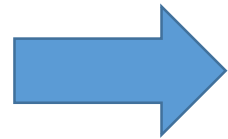
□ $V_{OH} = V_{DD}, V_{IH} = V_{DD}/2$

□ $NML = V_{IH} - V_{OL}$

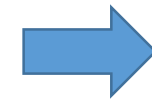
□ $V_{IH} = V_{DD}/2, V_{OL} = GND$



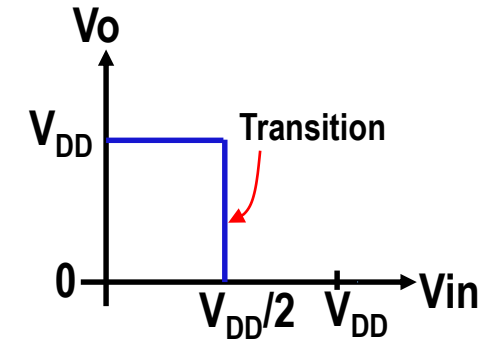
$NMH = V_{DD}/2$



$NML = V_{DD}/2$



$NMH = NML$



□ Practical CMOS inverter

□ To get V_{OH}

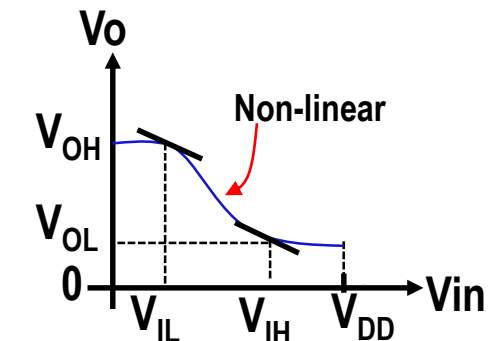
□ Apply $V_{in} = 0$, then find $V_o = V_{OH}$

□ To get V_{OL}

□ Apply $V_{in} = V_{DD}$, then find $V_o = V_{OL}$

□ To get V_{IH} and V_{IL}

□ Find relation between V_o and V_{in} , then substitute $\frac{dV_o}{dV_{in}} = -1$.

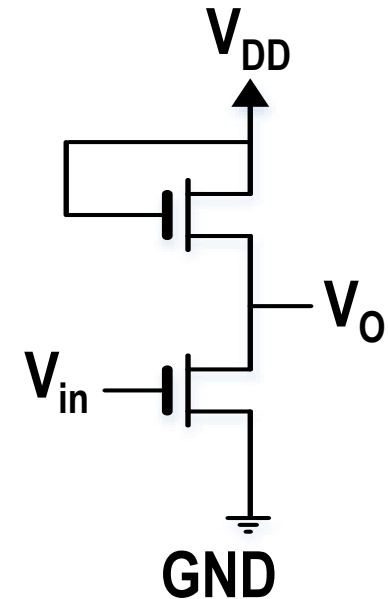
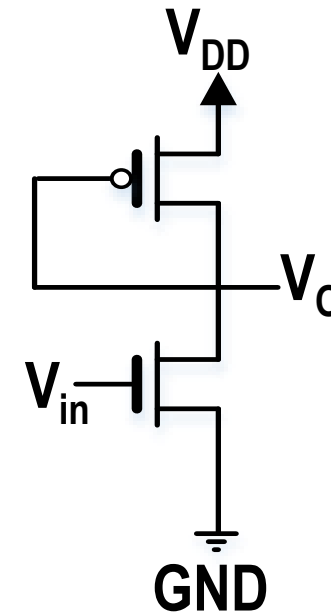
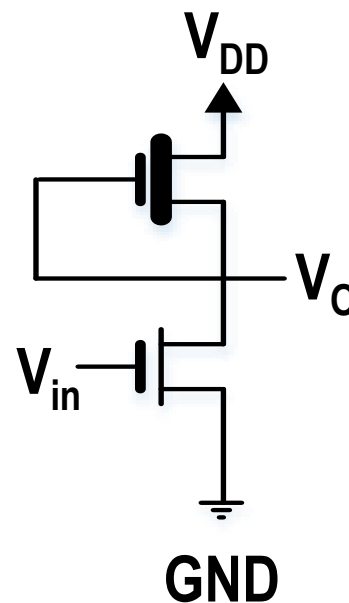
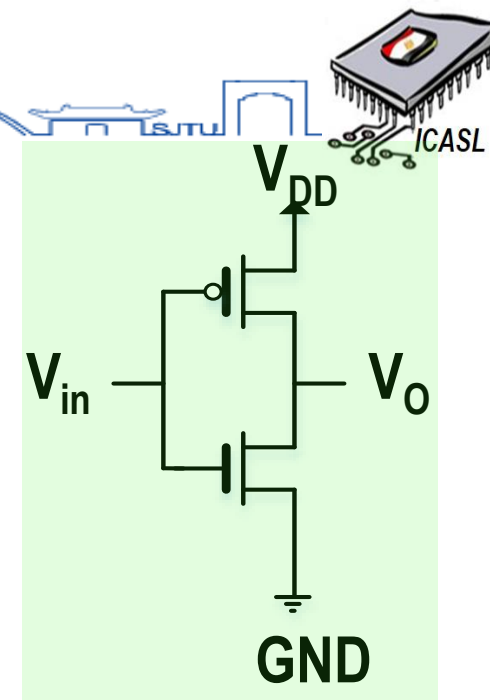
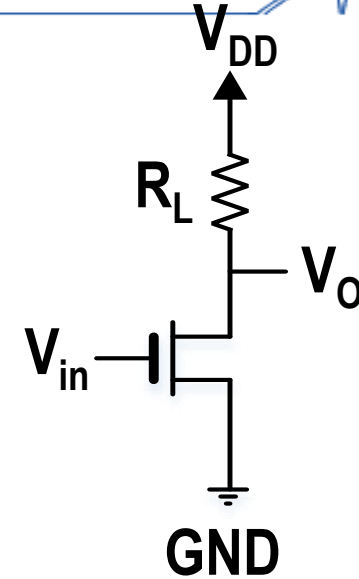
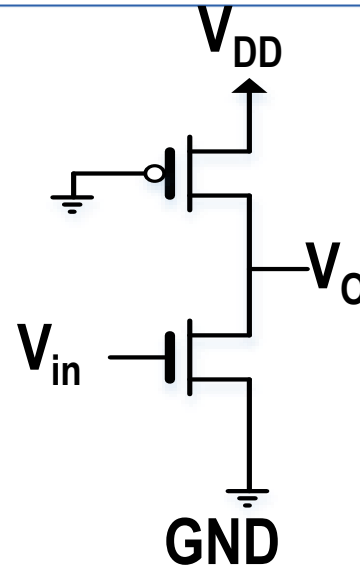




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- ☐ **MOS Inverter Topologies**
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- ☐ Discussion

MOS Inverter Topologies

- ☒ CMOS Inverter
- ☐ Static Load Inverter
- ☐ Pseudo (N-MOS) Inverter
- ☐ Saturation (N-MOS) Inverter
- ☐ More Saturation (N-MOS) Inverter
- ☐ Depletion (N-MOS) Inverter.



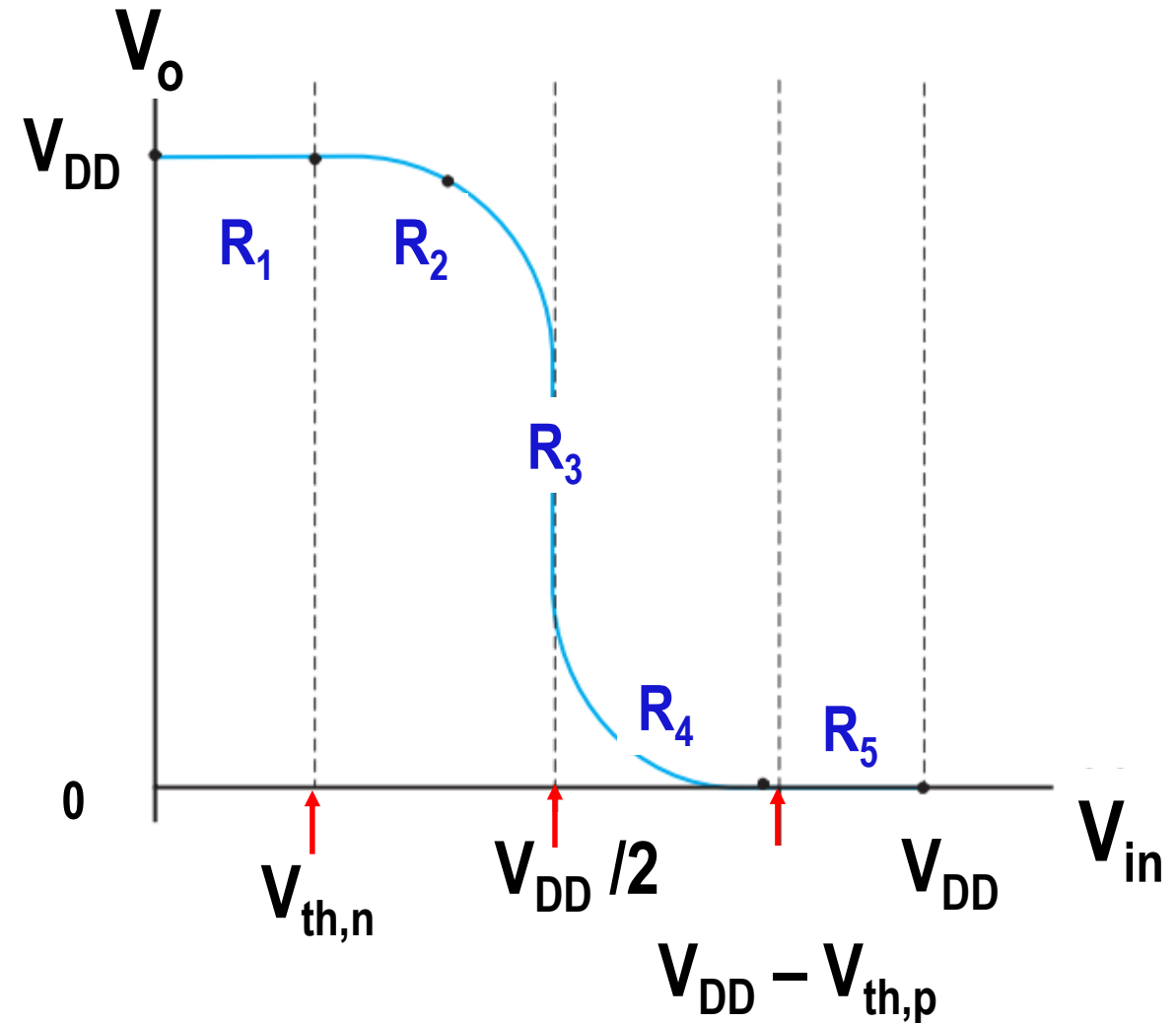
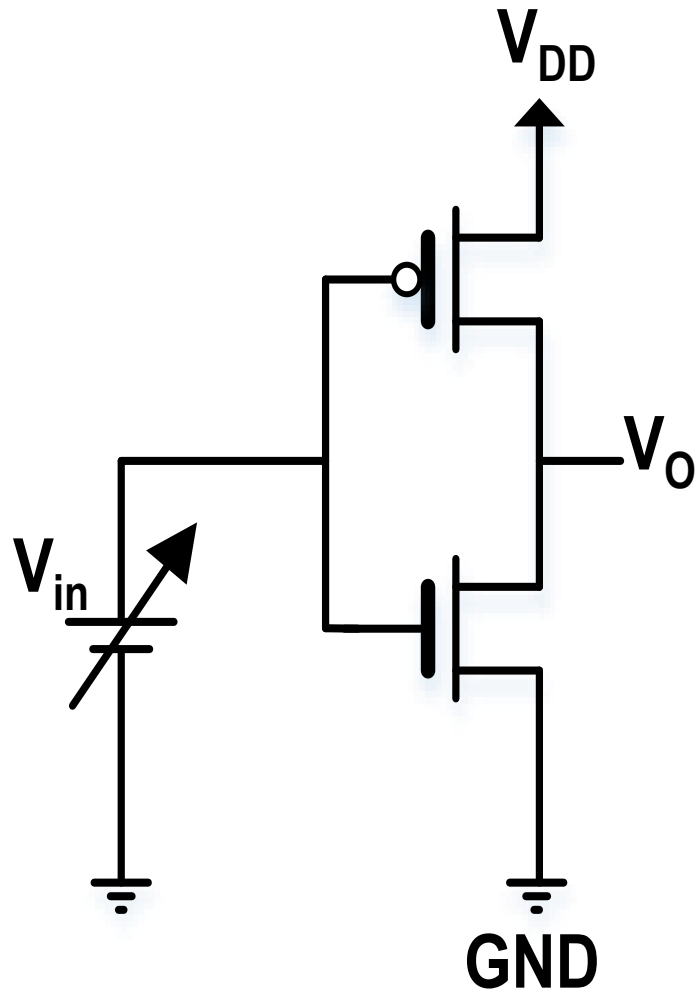


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DC Transfer Characteristics (1/8)

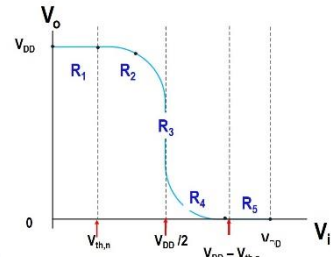
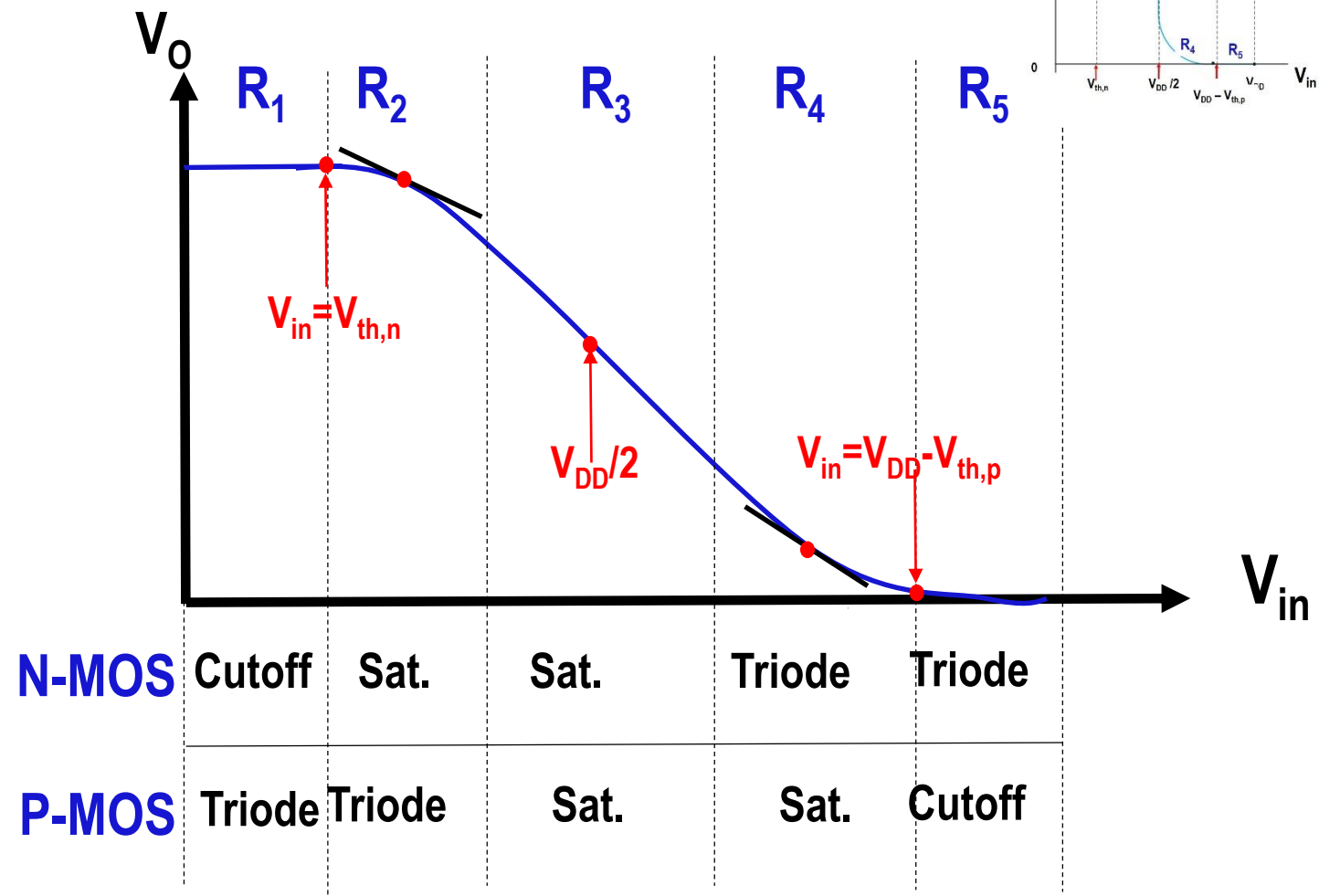
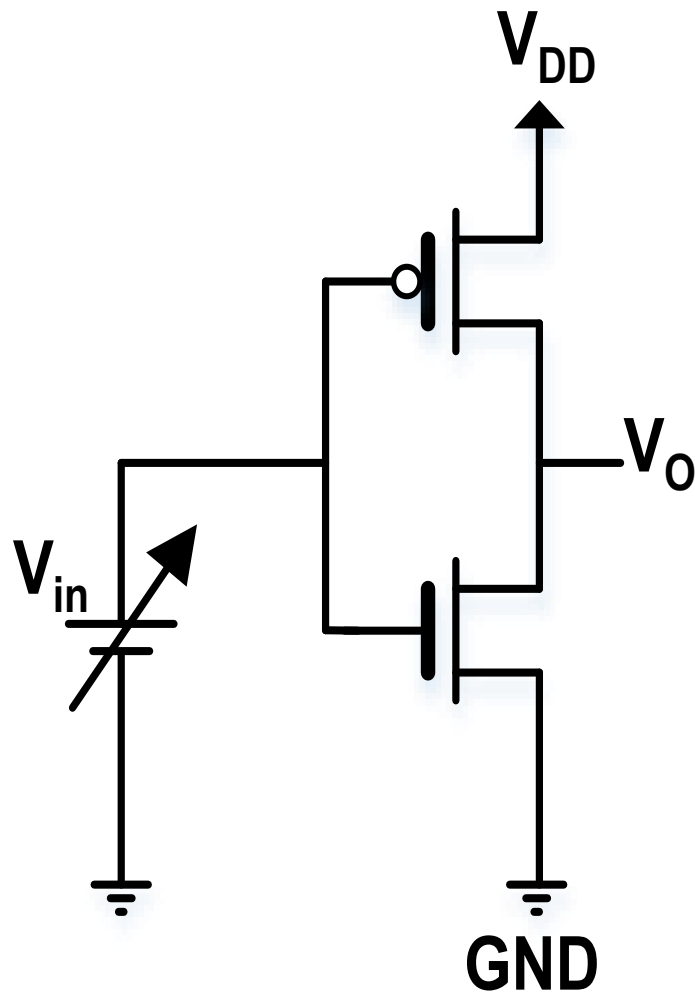


Static CMOS Inverter DC Characteristics



DC Transfer Characteristics (2/8)

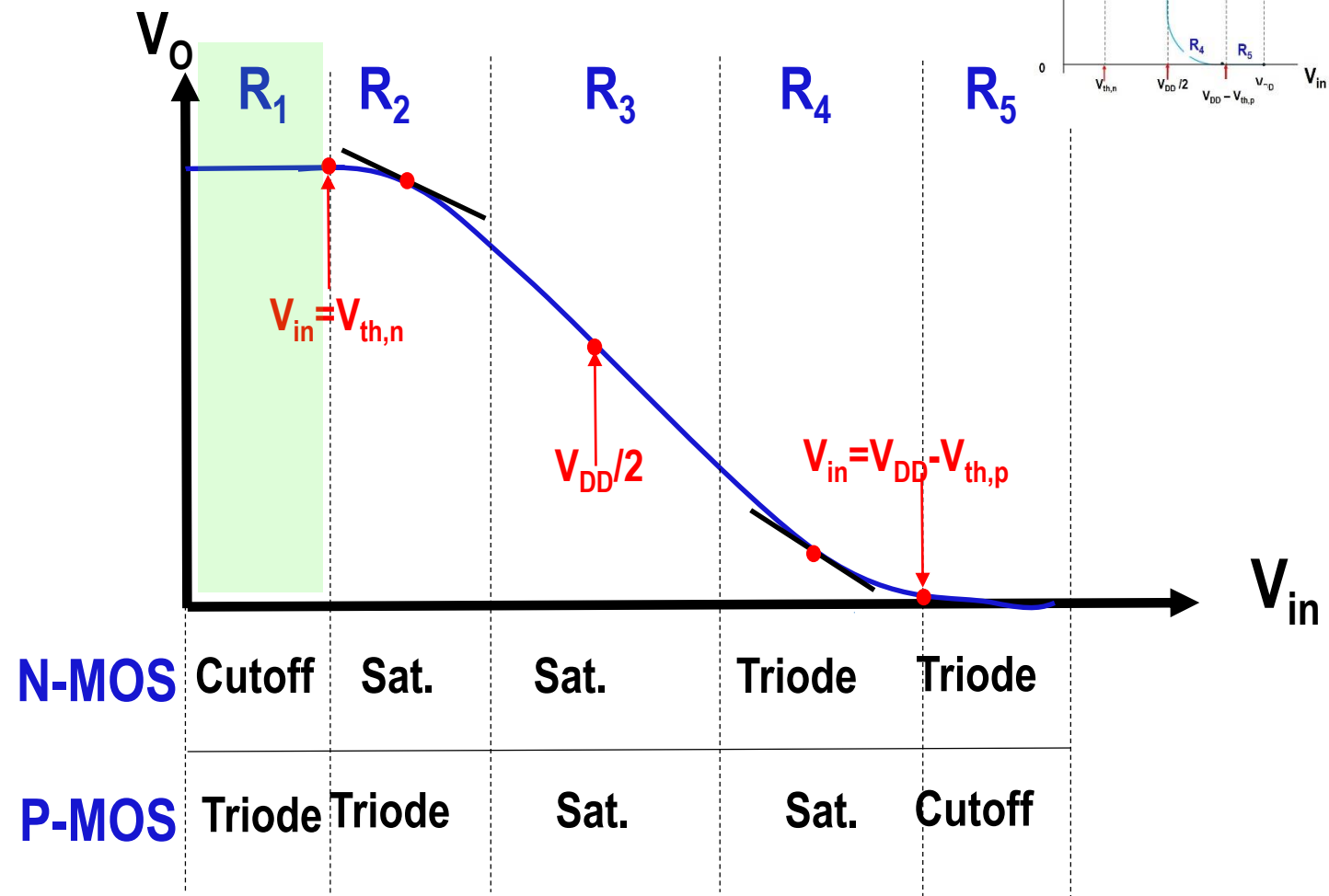
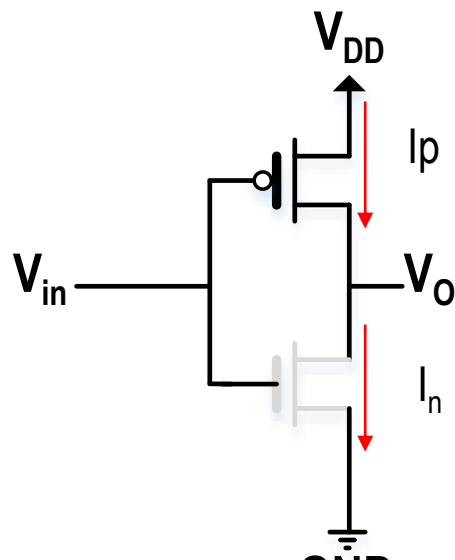
Static CMOS Inverter DC Characteristics



DC Transfer Characteristics (3/8)

Static CMOS Inverter DC Characteristics

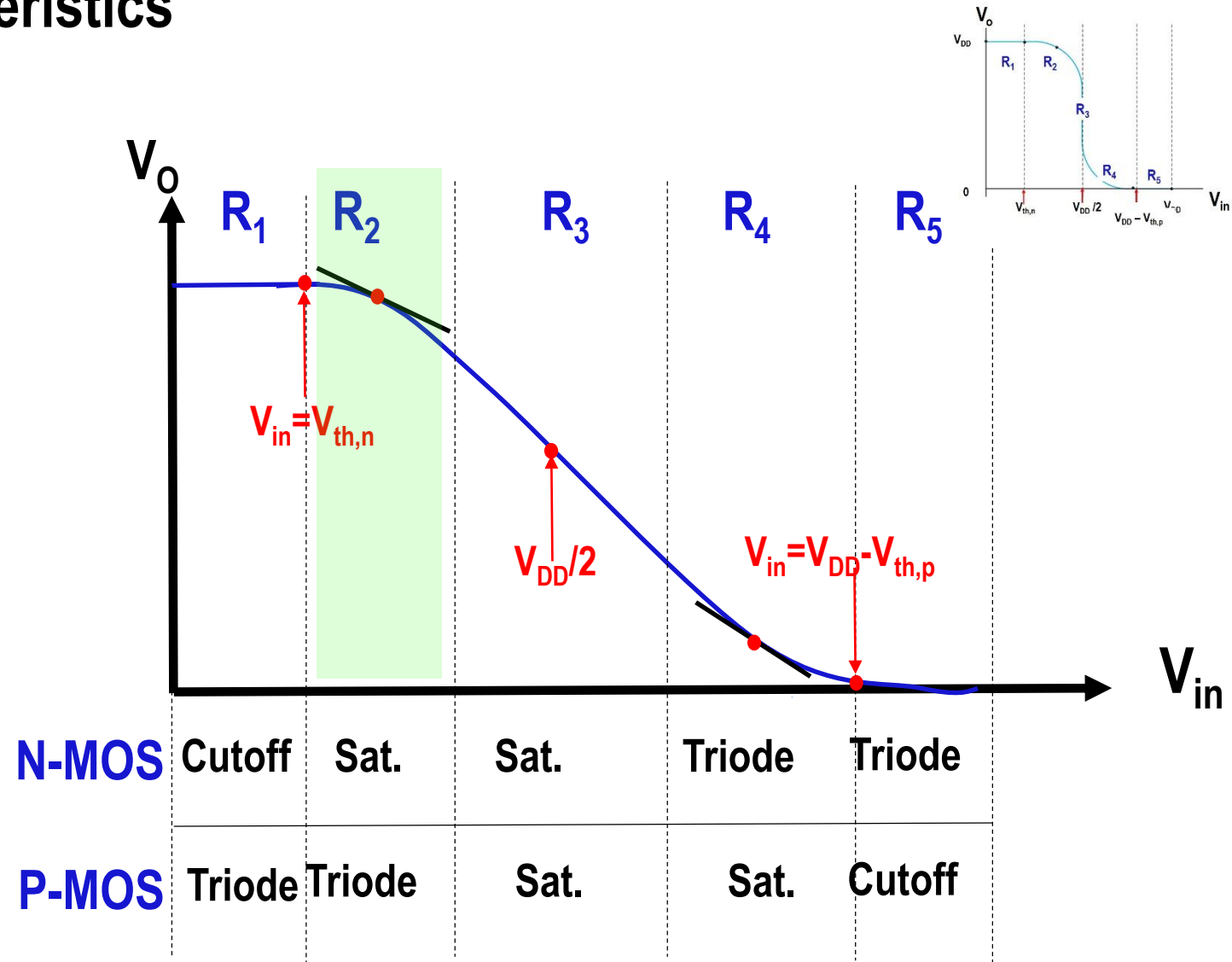
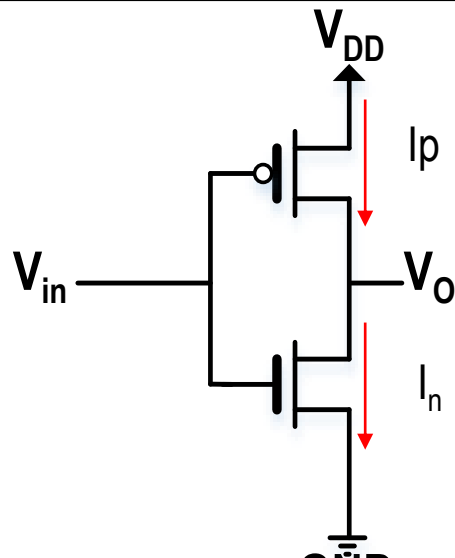
Region (1)	
$V_{in} < V_{th,n}$	
N-MOS	P-MOS
Cutoff (Open Circuit)	Triode
$V_O = V_{DD}$	
$I_{n(cutoff)} = 0 = I_{p(triode)}$	



DC Transfer Characteristics (4/8)

Static CMOS Inverter DC Characteristics

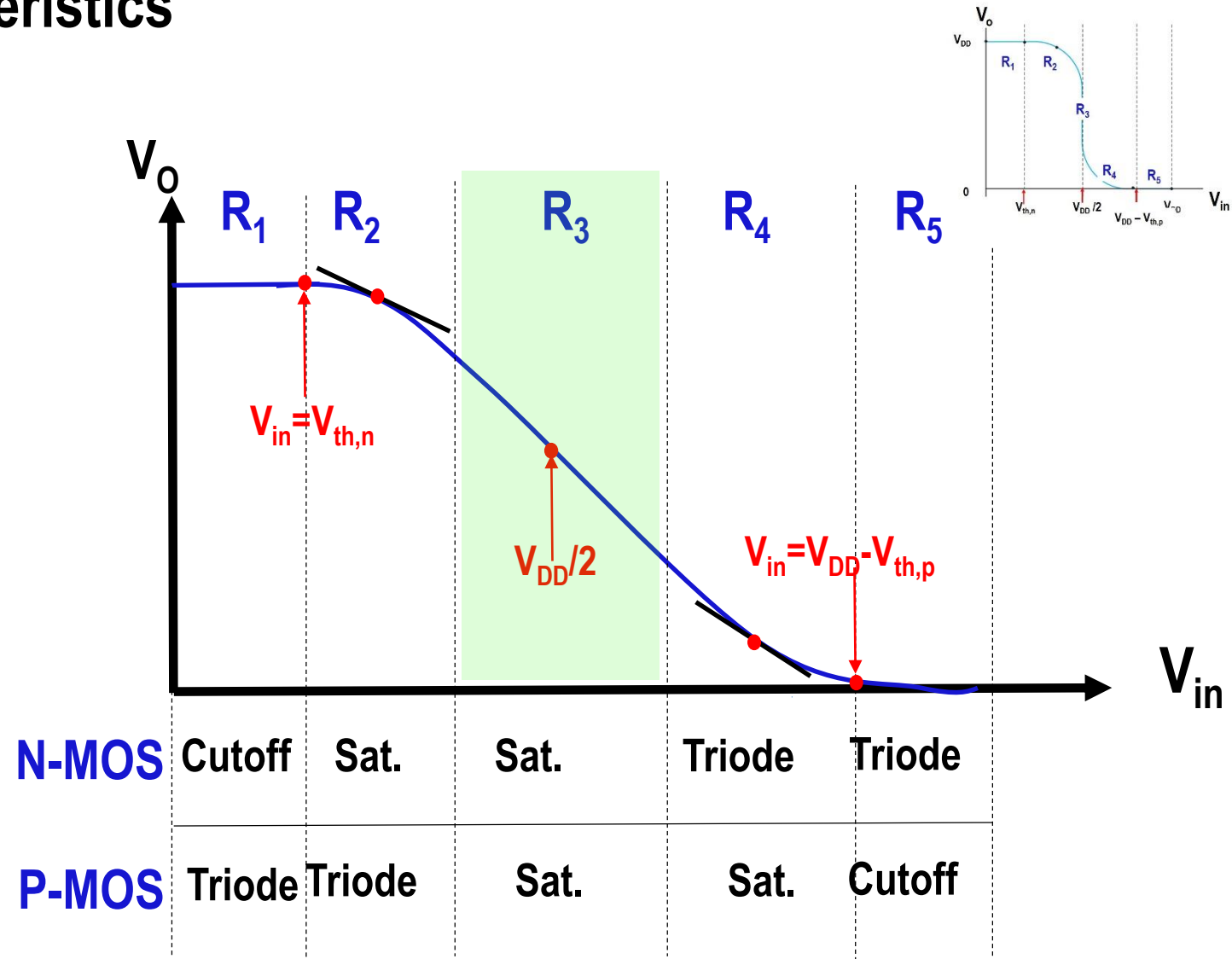
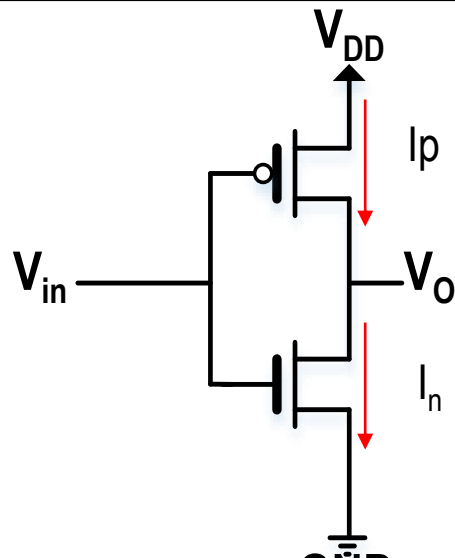
Region (2)	
$V_{DD}/2 > V_{in} \geq V_{th,n}$	
N-MOS	P-MOS
Saturation	Triode
$V_{ds,n} \geq V_{gs,n} - V_{th,n}$ $V_O \geq V_{in} - V_{th,n}$	$V_{ds,p} > V_{gs,p} - V_{th,p}$ $V_O > V_{in} - V_{th,p}$
$I_{n(Sat.)} = I_{p(triode)}$	



DC Transfer Characteristics (5/8)

Static CMOS Inverter DC Characteristics

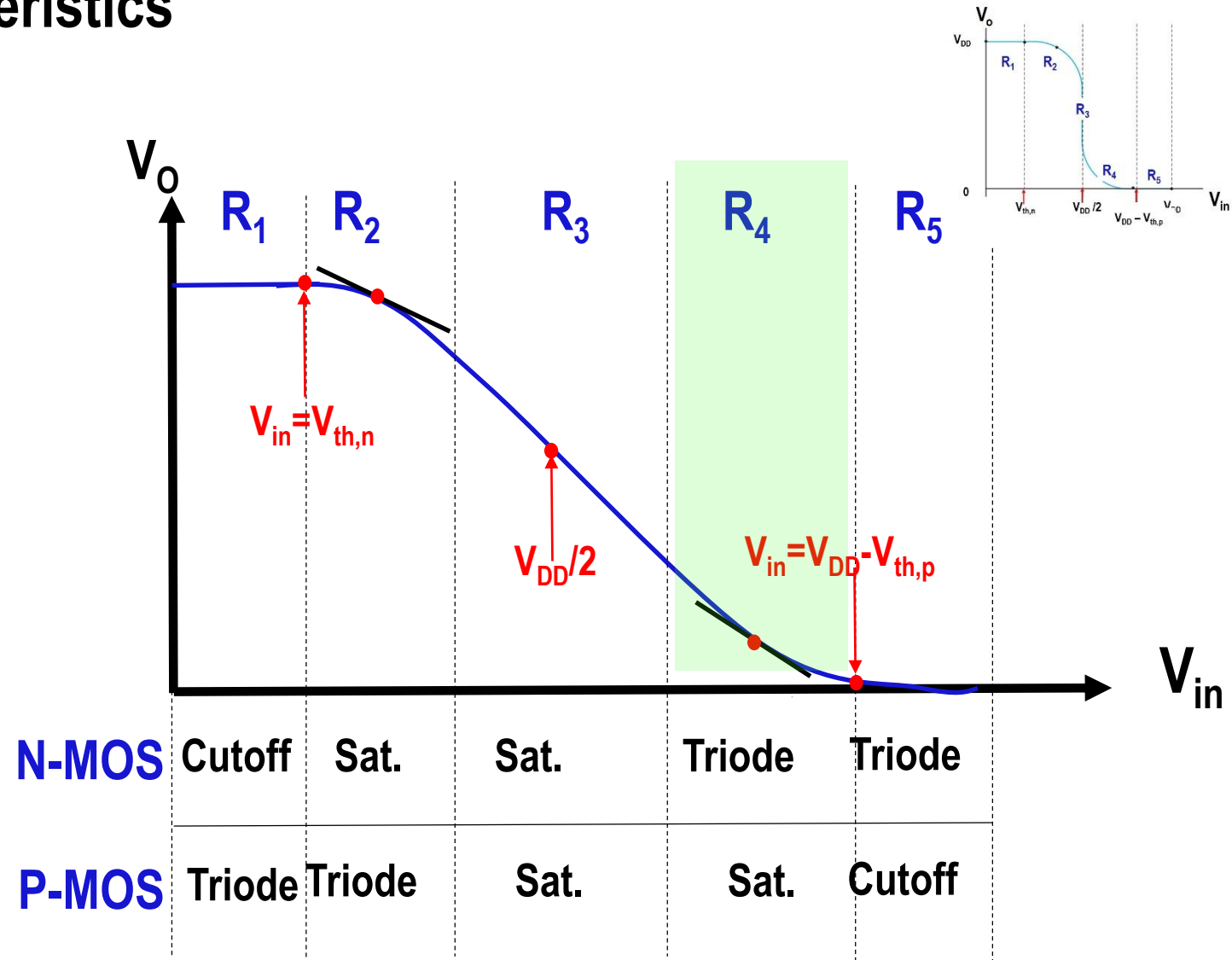
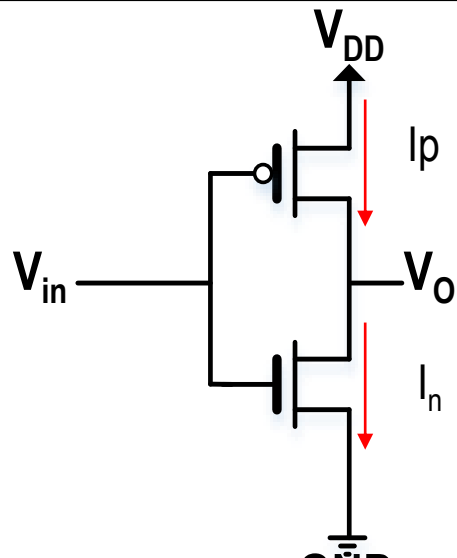
Region (3)	
$V_{in} = V_{DD}/2$	
N-MOS	P-MOS
Saturation	Saturation
$V_{ds,n} \geq V_{gs,n} - V_{th,n}$ $V_O \geq V_{in} - V_{th,n}$	$V_{ds,p} \leq V_{gs,p} - V_{th,p}$ $V_O \leq V_{in} - V_{th,p}$
$I_{n(Sat.)} = I_{p(Sat.)}$	



DC Transfer Characteristics (6/8)

Static CMOS Inverter DC Characteristics

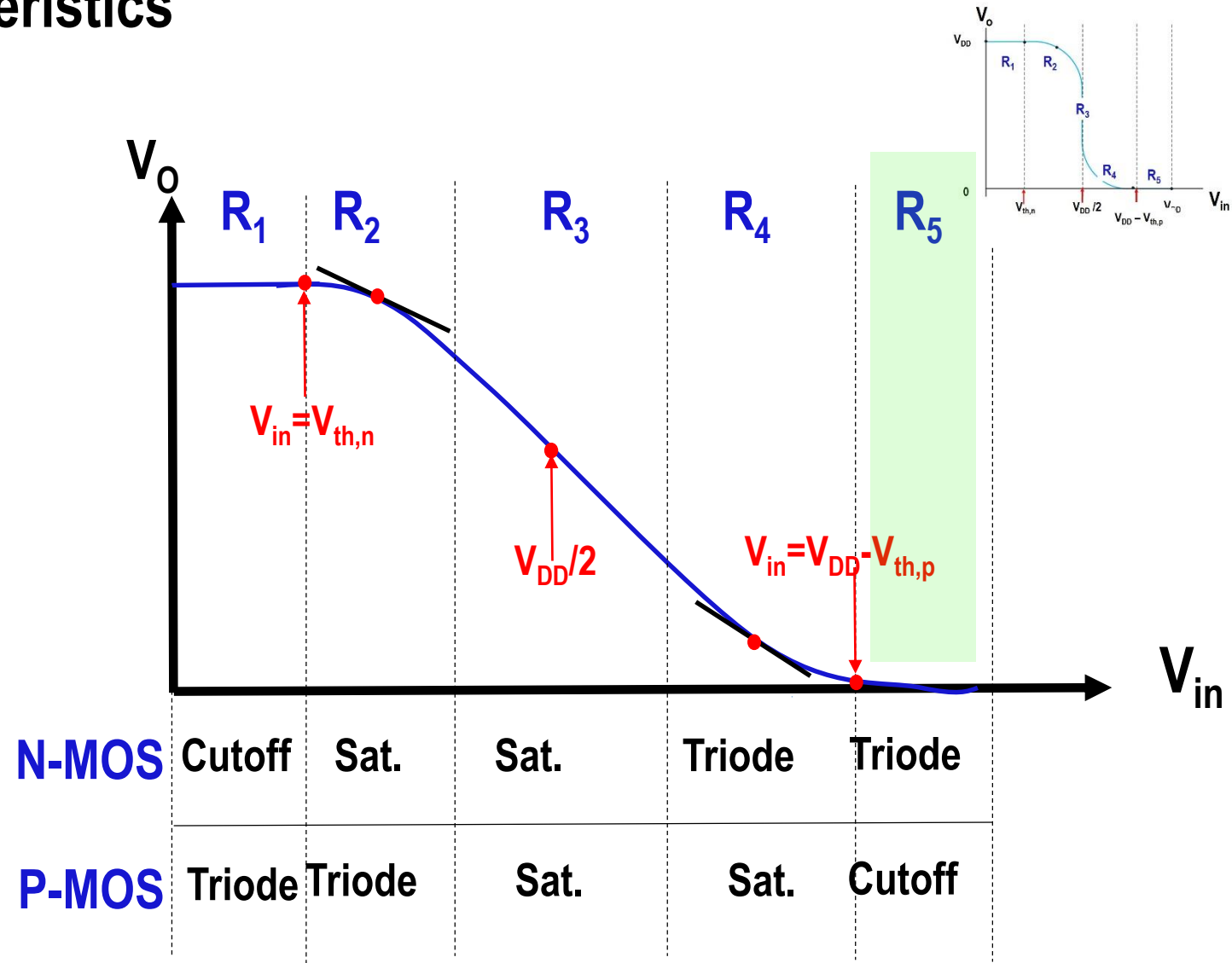
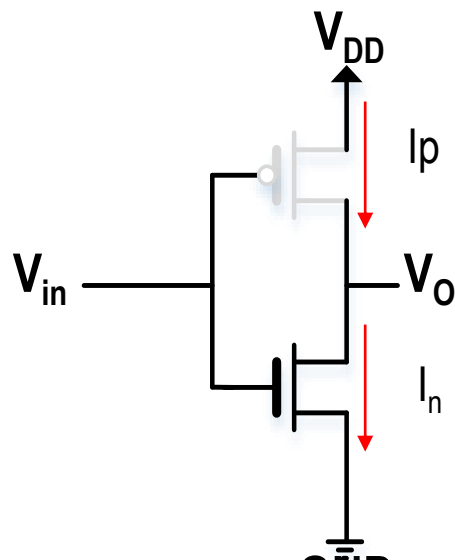
Region (4)	
$V_{DD} - V_{th,p} > V_{in} \geq V_{DD}/2$	
N-MOS	P-MOS
Triode	Saturation
$V_{ds,n} < V_{gs,n} - V_{th,n}$ $V_O < V_{in} - V_{th,n}$	$V_{ds,p} \leq V_{gs,p} - V_{th,p}$ $V_O \leq V_{in} - V_{th,p}$
$I_{n(Triode)} = I_{p(Sat.)}$	



DC Transfer Characteristics (7/8)

Static CMOS Inverter DC Characteristics

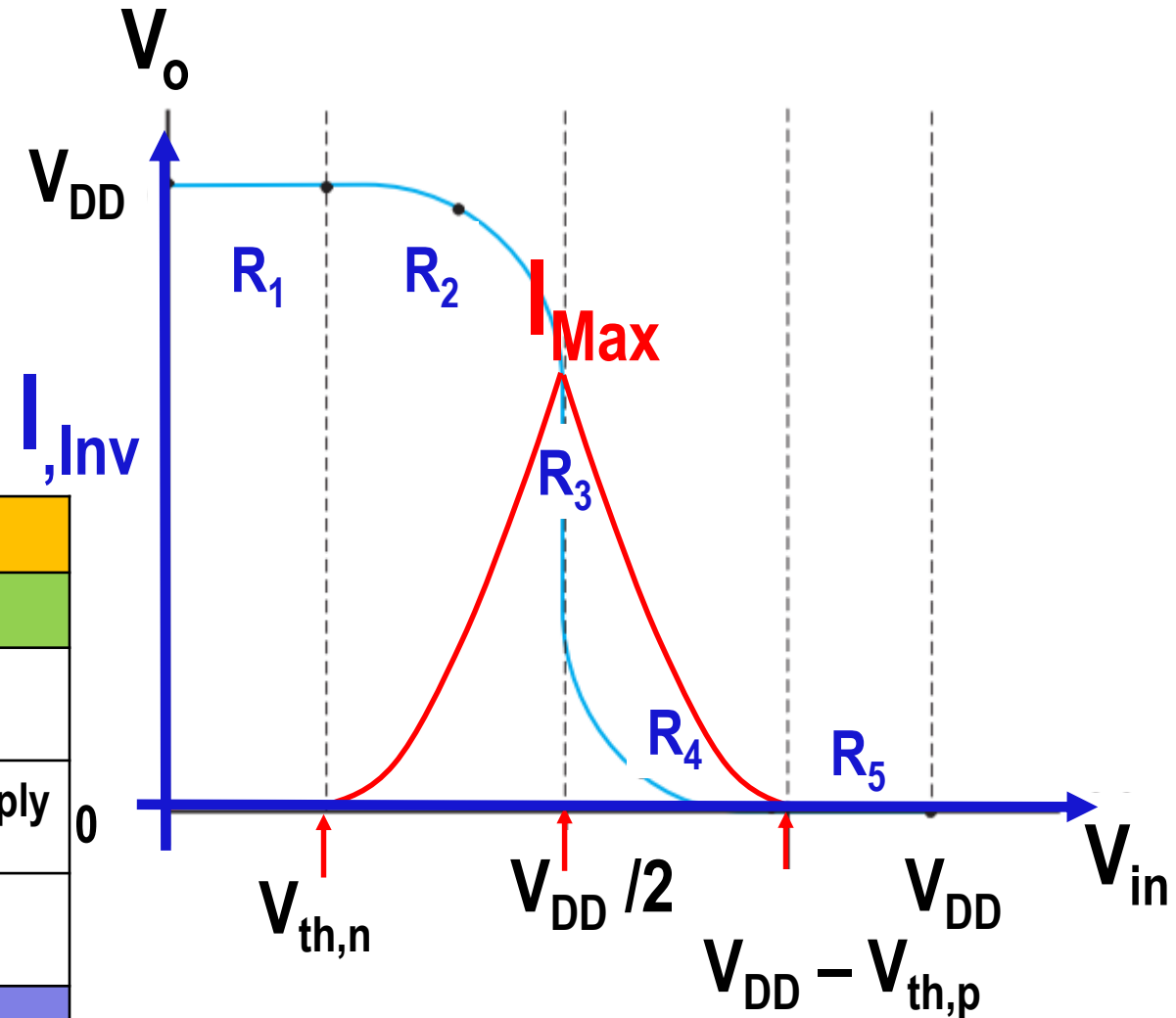
Region (5)	
$V_{in} \geq V_{DD} - V_{th,p}$	
N-MOS	P-MOS
Triode	Cutoff
$V_o = 0$	
$I_{p(Cutoff)} = 0 = I_{n(Triode)}$	



DC Transfer Characteristics (8/8)

- ❑ **Maximum** current will be occurred
When both N-MOS and P-MOS are
operating in saturation region (region 3)

Region	Condition	N-MOS	P-MOS	Output
1	$0 < V_{in} < V_{th,n}$	Cutoff	Triode	$V_o = V_{DD}$
2	$V_{th,n} \leq V_{in} < V_{DD}/2$	Sat.	Triode	$V_o > V_{DD}/2$
3	$V_{in} = V_{DD}/2$	Sat.	Sat.	V_o drops sharply
4	$V_{DD}/2 < V_{in} \leq V_{DD} - V_{th,p}$	Triode	Sat.	$V_o < V_{DD}/2$
5	$V_{in} \geq V_{DD} - V_{th,p}$	Triode	Cutoff	$V_o = 0$





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Beta Ratio Effects (1/3)



$$B_n = \mu_n C_{ox} \frac{W_n}{L_n}$$

B_n : Gain process factor of n-MOSFET

μ_n : Mobility of Electron

C_{ox} : Gate Oxide Capacitor/unit area

W_n : width of transistor

L_n : length of transistor

$$B_p = \mu_p C_{ox} \frac{W_p}{L_p}$$

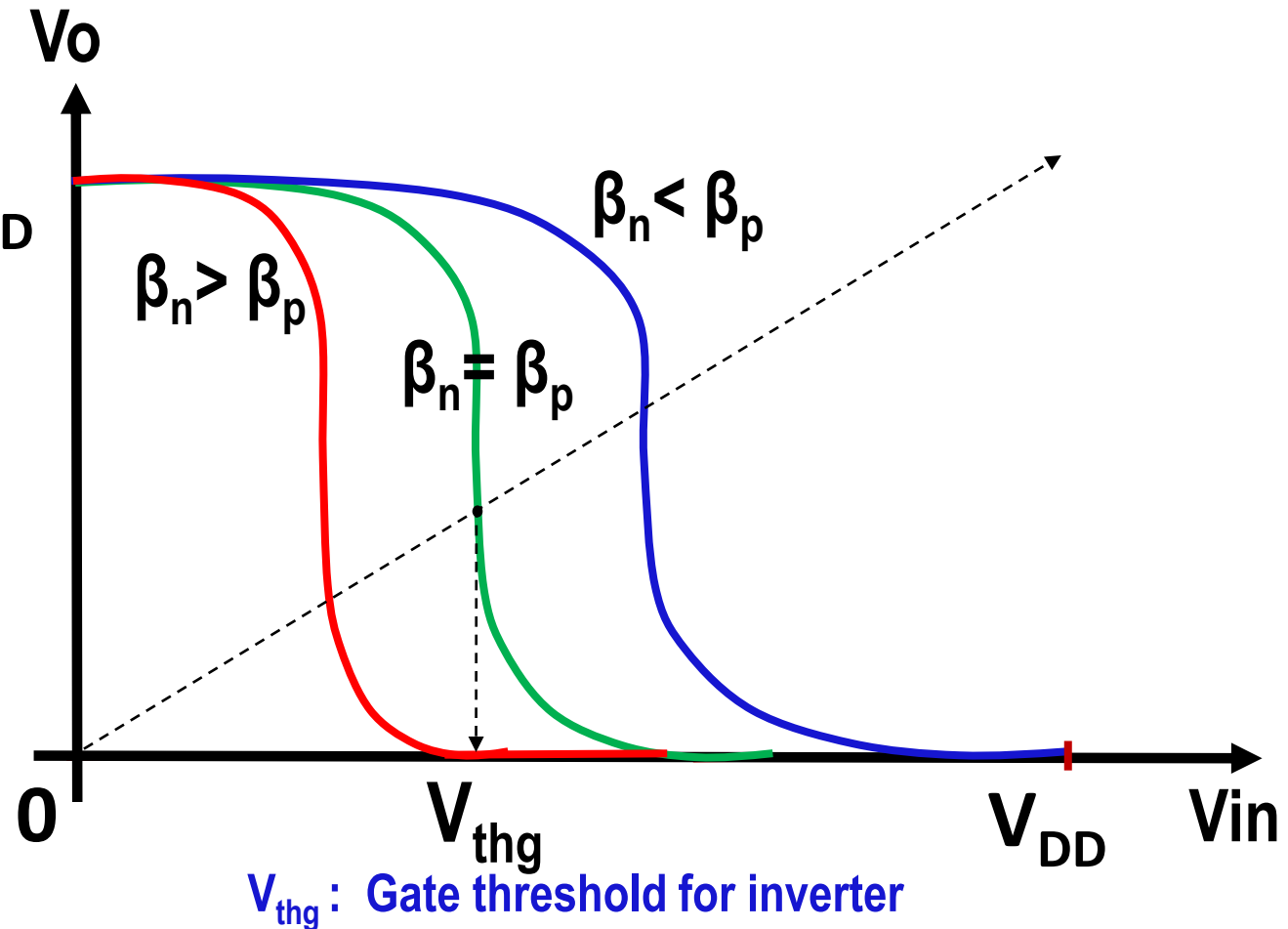
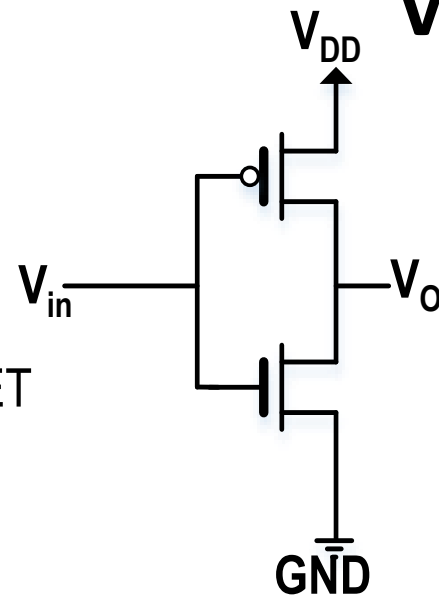
B_p : Gain process factor of p-MOSFET

μ_p : Mobility of Hole

C_{ox} : Gate Oxide Capacitor/unit area

W_p : width of transistor of p-MOSFET

L_p : length of transistor of p-MOSFET



$$V_{thg} = \frac{V_{th,n} + r(V_{DD} + V_{th,p})}{1 + r}$$

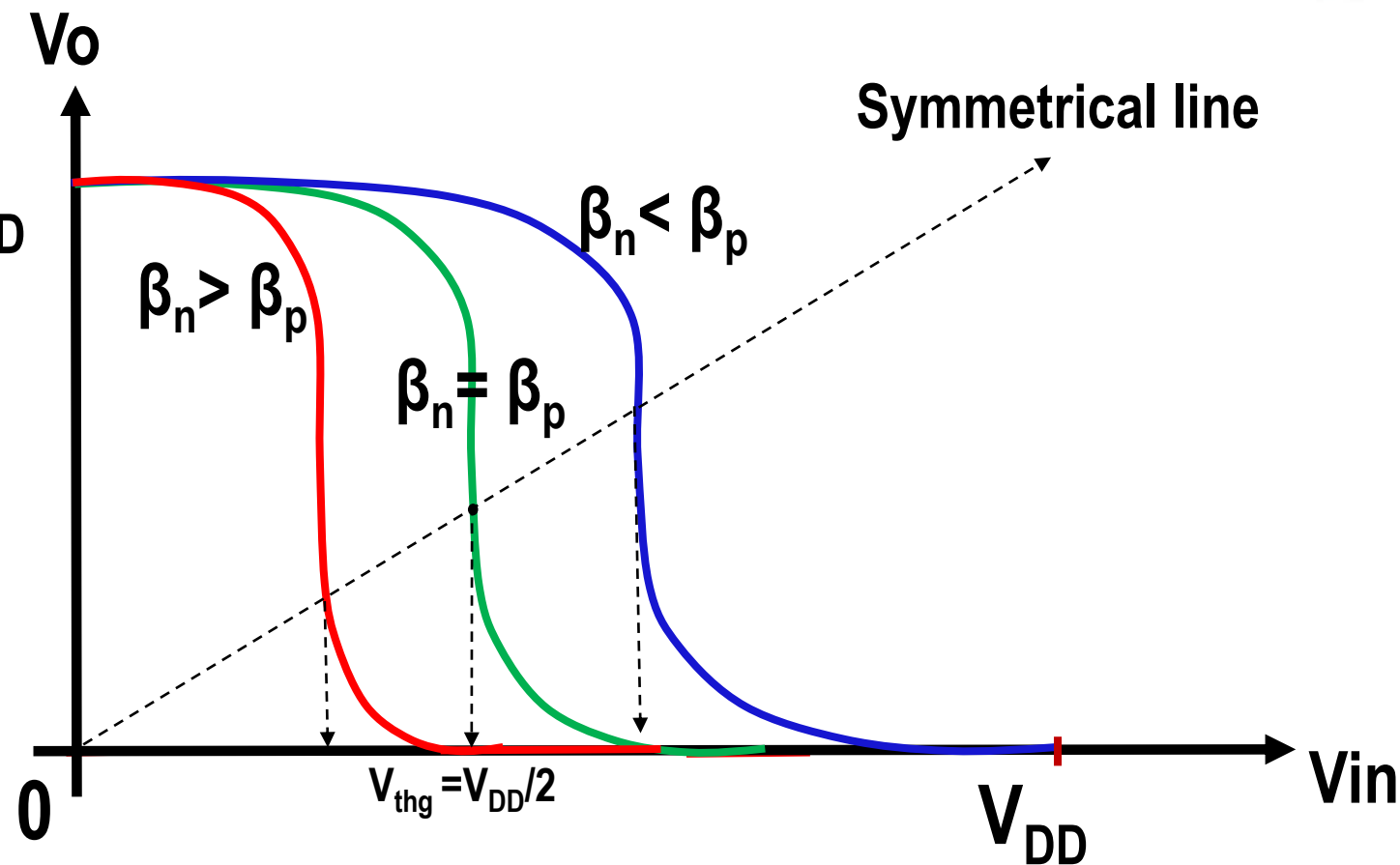
$$\text{Where } r = \sqrt{\frac{\beta_p}{\beta_n}}$$

Beta Ratio Effects (2/3)



Case	Condition	V_{thg}
1	$\beta_n = \beta_p$, $V_{th,n} = V_{th,p} $	$V_{thg} = V_{DD}/2$
2	$\beta_n > \beta_p$, $V_{th,n} = V_{th,p} $	$V_{thg} < V_{DD}/2$
3	$\beta_n < \beta_p$, $V_{th,n} = V_{th,p} $	$V_{thg} > V_{DD}/2$

Inverters with different beta are called *skewed inverters*



To obtain $\beta_n = \beta_p$, $W_p = 2:3 W_n$ & $L_p = L_n$ because $\mu_n = 2:3 \mu_p$

To obtain $\beta_n = \beta_p$, $W_p = 2.5 W_n$ & $L_p = L_n$ because $\mu_n = 2.5 \mu_p$

Symmetrical
Condition

Beta Ratio Effects (3/3)

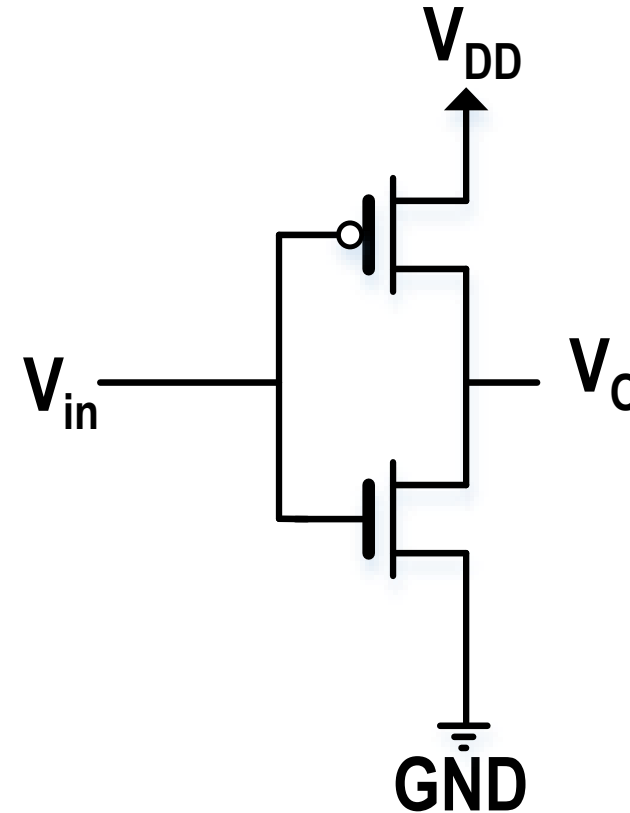


□ Symmetrical condition

□ $\beta_n = \beta_p$, $W_p = 2.5 W_n$ & $L_p = L_n$

□ Minimum Silicon Area

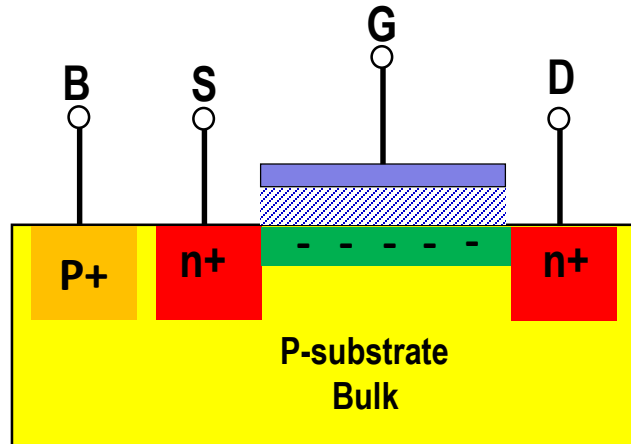
□ $\beta_n > \beta_p$, $W_p = W_n$ & $L_p = L_n$



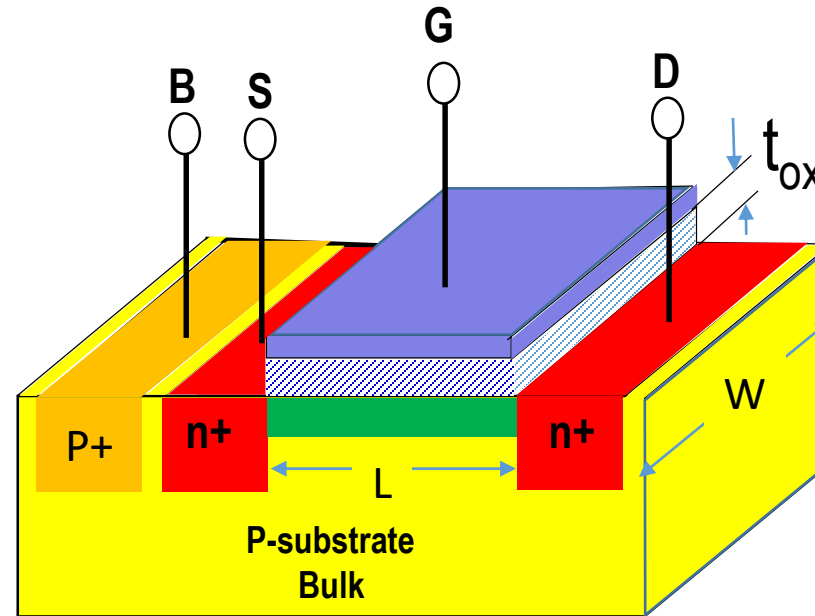


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Dynamic Characteristics of CMOS Inverter (1/2)



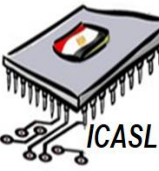
2-D n-MOSFET



3-D n-MOSFET

The most important element that impacts **strongly** the dynamic characteristics of CMOS inverter is the **capacitor**.

Dynamic Characteristics of CMOS Inverter (2/2)



□ Total Capacitance of CMOS Inverter

□ Assume we have two connected CMOS inverters as shown.

$$C_{\text{total}} = C_{\text{in}} + C_{\text{o}} + C_{\text{r}}$$

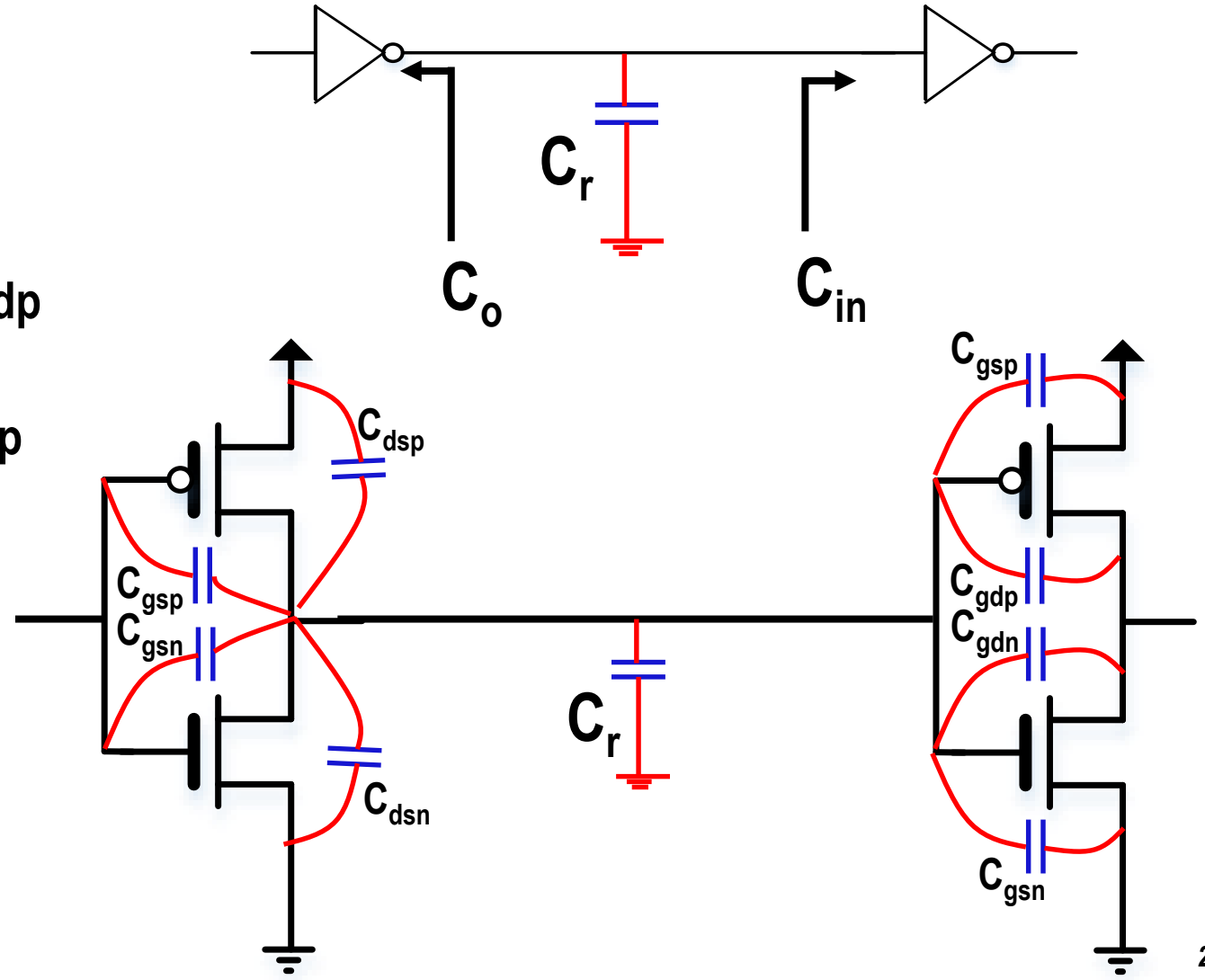
$$C_{\text{in}} = C_{\text{gsn}} + C_{\text{gsp}} + 2 * C_{\text{gdn}} + 2 * C_{\text{gdp}}$$

$$C_{\text{o}} = C_{\text{dsn}} + C_{\text{dsp}} + 2 * C_{\text{gdn}} + 2 * C_{\text{gdp}}$$

C_{r} : routing capacitance

C_{o} : output capacitance

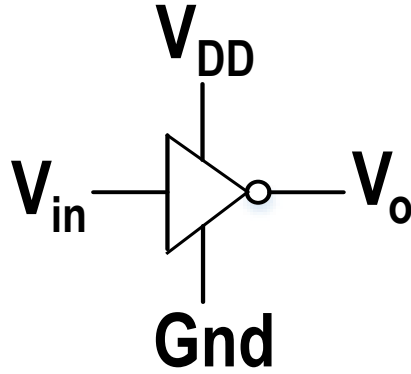
C_{in} : input capacitance





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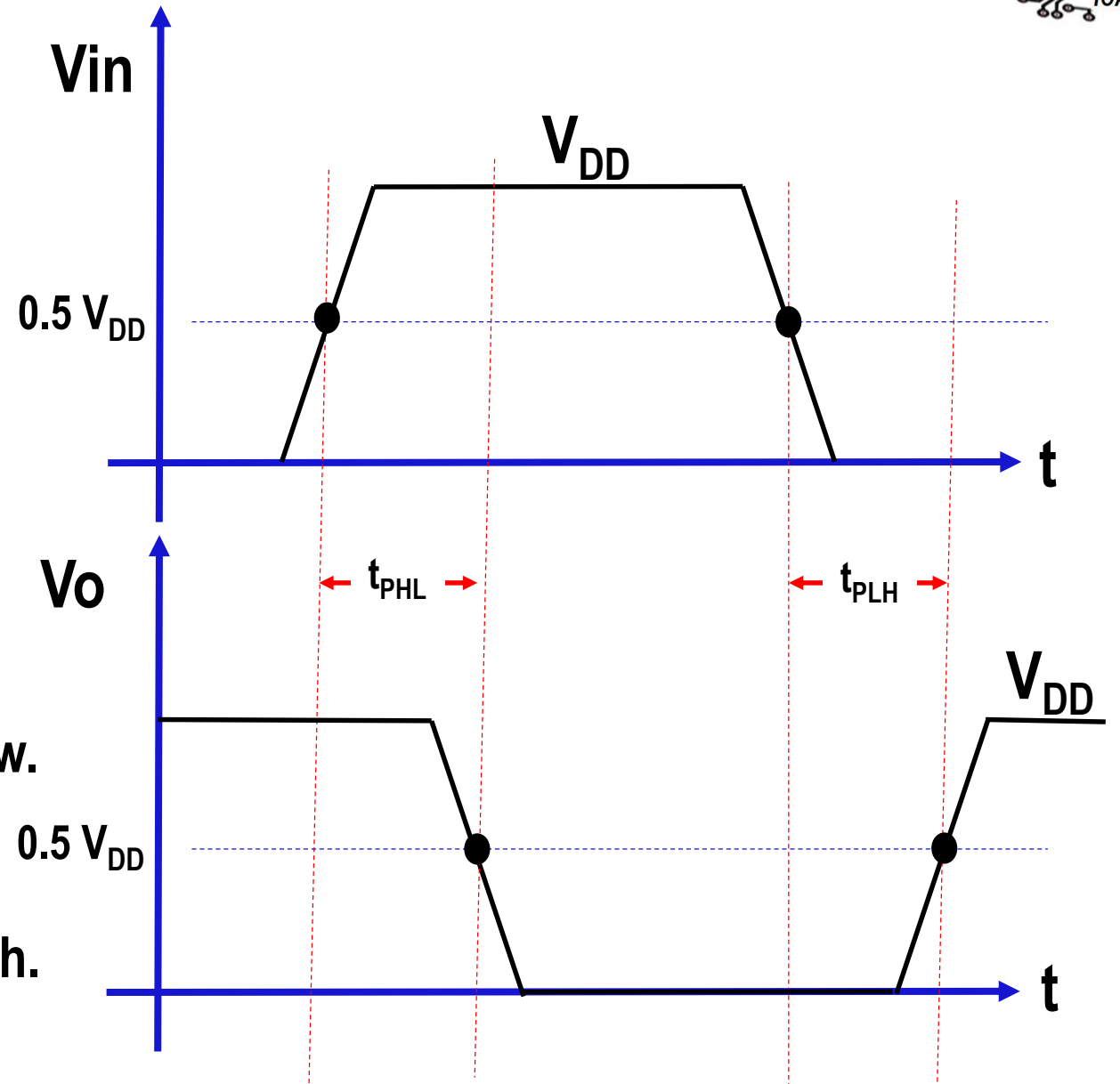
Inverter switching characteristics (1/3)



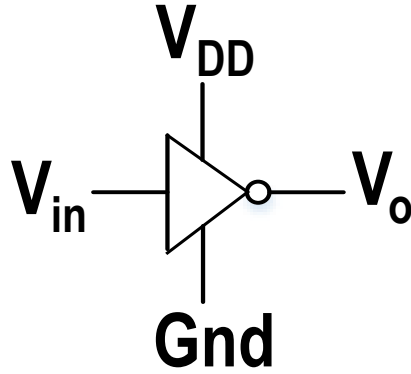
- **Propagation time (t_p):** is the time between 50 % of the input voltage to 50 % of the output.

$$t_p = \frac{(t_{PHL} + t_{PLH})}{2}$$

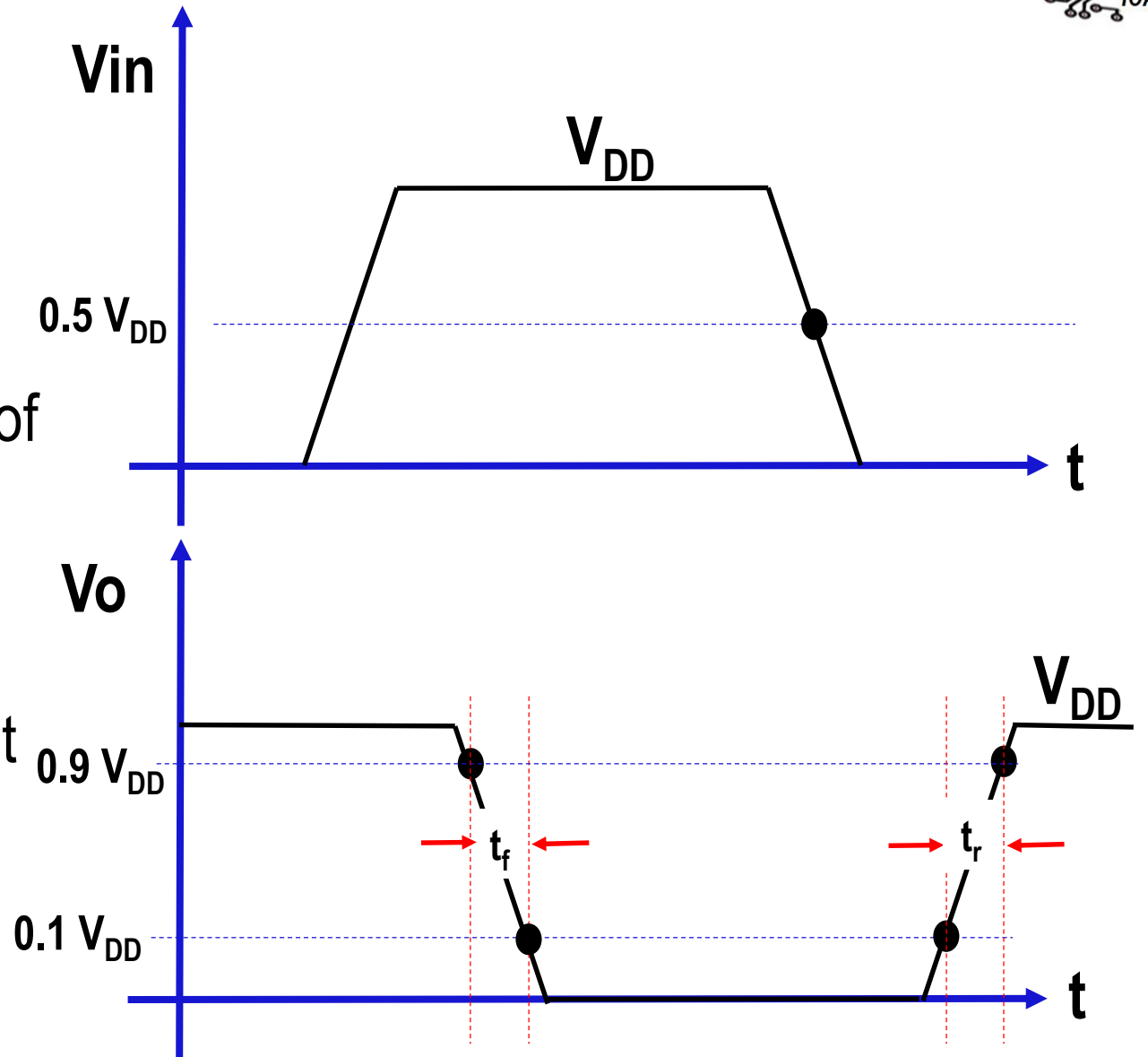
- t_{PHL} : is the propagation time from high to low.
- t_{PLH} : is the propagation time from low to high.



Inverter switching characteristics (2/3)



- ❑ **Rise time (t_r)**: is the time between 10% of the output voltage to 90 % of the output.
- ❑ **Fall time (t_f)**: is the time between 90% of the output voltage to 10% of the output



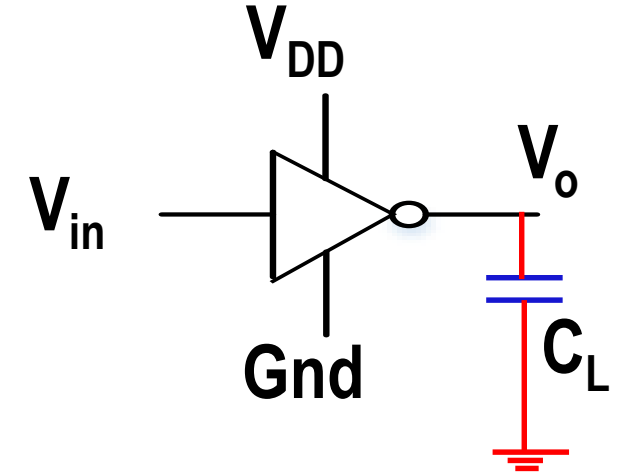
Inverter switching characteristics (3/3)



When, $\beta_n = \beta_p = \beta$ and $V_{th,n} = V_{th,p} = 0.2 * V_{DD}$

$$t_p = \frac{1.61 * C_L}{\beta * V_{DD}}$$

$$t_r = t_f = \frac{3.67 * C_L}{\beta * V_{DD}}$$



Propagation time, rising time and falling time are dependent on

- ❑ Load capacitance C_L
- ❑ Power supply V_{DD}
- ❑ Gain process factor β

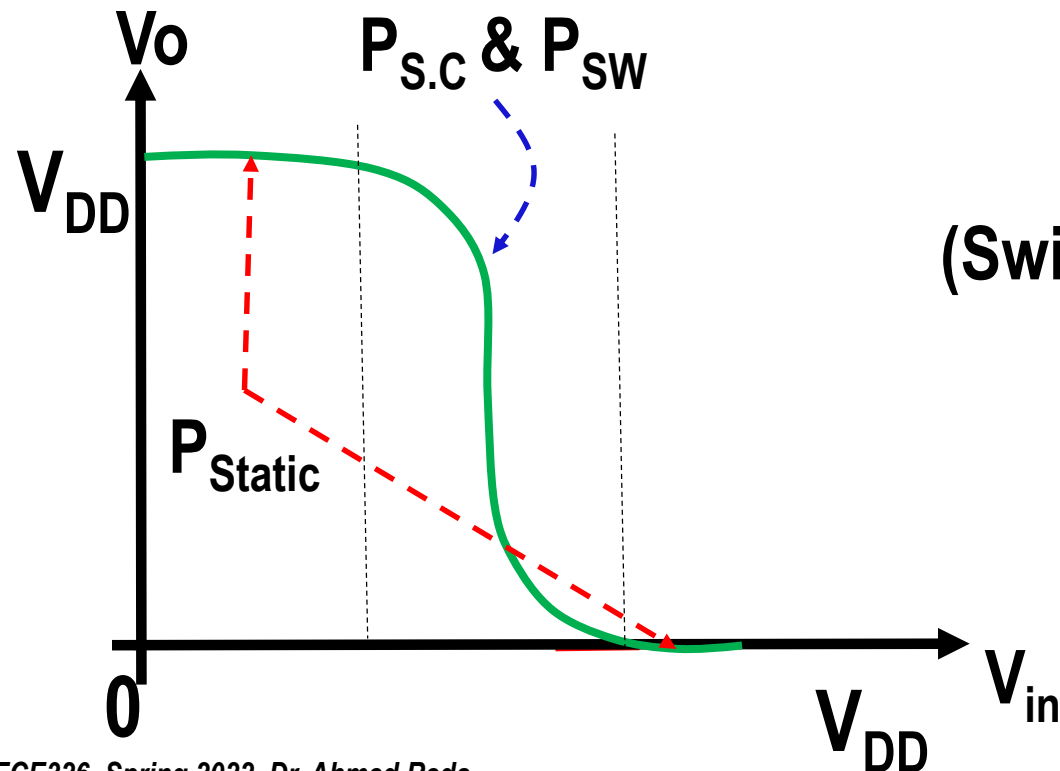


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Power Dissipation

Static power dissipation (P_{static})

Dynamic power dissipation



Charging/Discharging

(Switching) Power Dissipation

(P_{sw})

Short Circuit

Power Dissipation

($P_{S.C}$)

$$P_{\text{Total}} = P_{\text{Static}} + P_{\text{Dynamic}}$$

$$P_{\text{Total}} = P_{\text{Static}} + P_{sw} + P_{S.C}$$

Static Power Dissipation

Definition:

Is the consumed power when passing a constant current due to apply a constant voltage.

Source of P_{static} .

Sub-threshold leakage

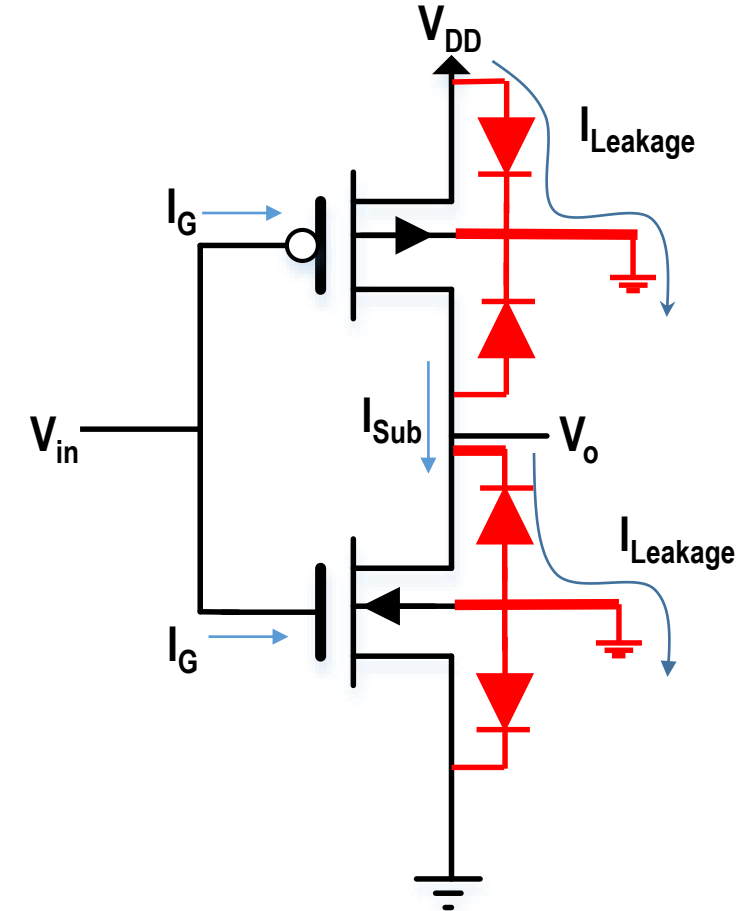
When $V_{gs,n} < V_{th,n} \rightarrow$ there is a very small current ($I_{leakage}$) in range of Pico-Ampere will pass through the transistor

Gate-leakage

Current would pass from gate to the channel $\rightarrow I_G \neq 0$

Junction-leakage

Current would pass through the parasitic diodes due to junction

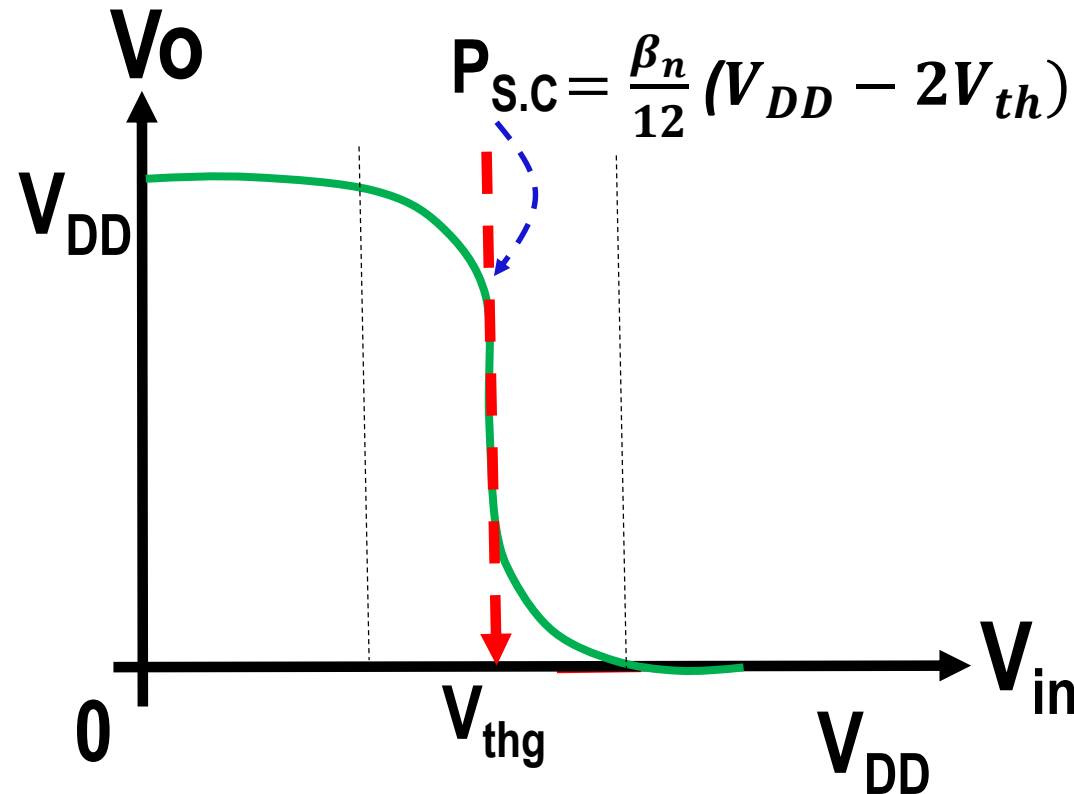
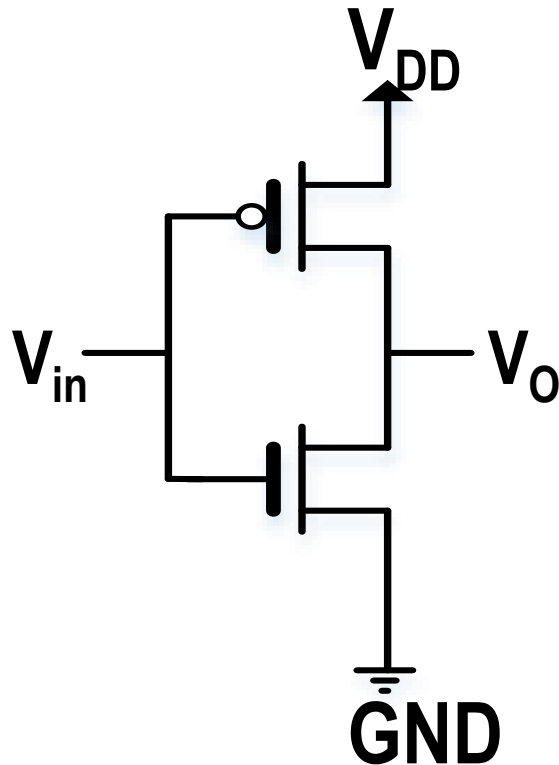


$$P_{Static} = (I_{sub} + I_G + I_{Junction}) * V_{DD}$$

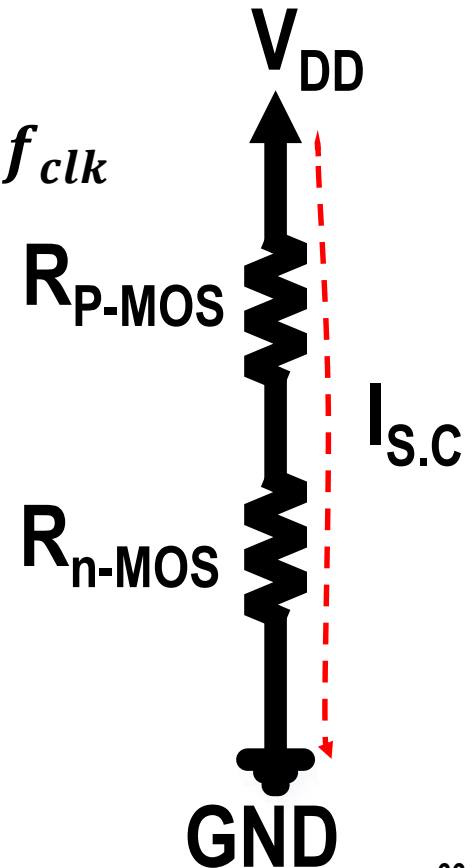
Shot circuit power dissipation

□ Definition:

Is the consumed power within the transient region. It is called power of short circuit because the current will pass from V_{DD} to GND while the N-MOS and P-MOS are operating "ON".



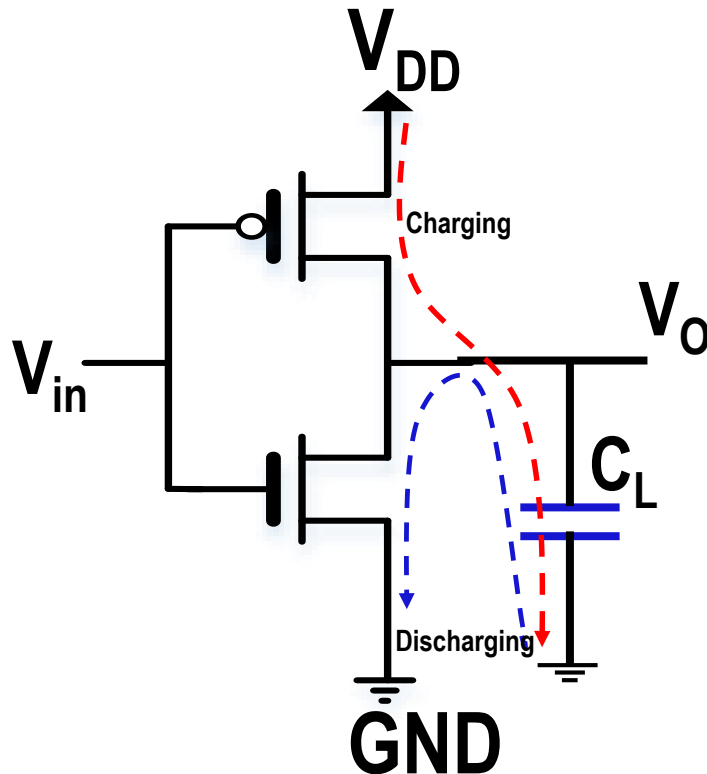
$$P_{s.c} = \frac{\beta_n}{12} (V_{DD} - 2V_{th})^3 * t_r * f_{clk}$$



Switching Power Dissipation “Charging/discharging”

□ Definition:

Is the consumed power during charging and discharging the capacitor



$$P_{SW} = C_L * V_{DD}^2 * f_{sw}$$
$$= \alpha * C_L * V_{DD}^2 * f_{clk}$$

f_{sw} : Switching frequency

f_{clk} : System clock frequency

$$f_{sw} = \alpha * f_{clk}$$

α : Activity factor ($0 < \alpha < 1$)

Note

$$P_{SW} \gg P_{S.C}$$



- ☐ DC transfer Ch/CS Curve
- ☐ Noise Margin Concept
- ☐ MOS Inverter Topologies
- ☐ DC Transfer Characteristics
- ☐ Beta Ratio Effects
- ☐ Dynamic Characteristics of CMOS Inverter
- ☐ Inverter switching characteristics
- ☐ Power Dissipation
- ☐ Discussion

Discussion (1/4)



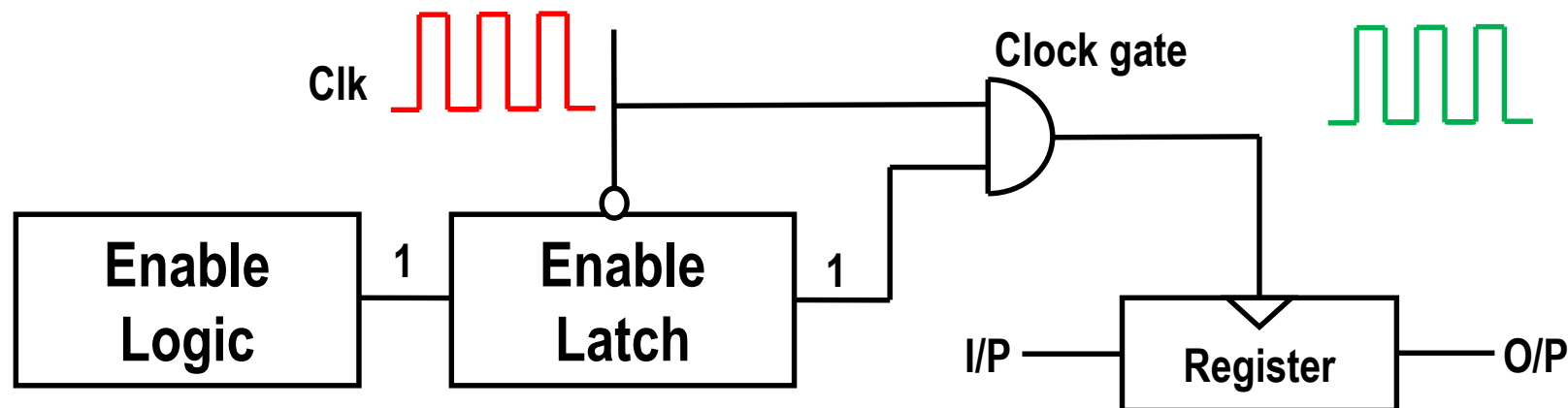
❑ What are the dynamic power reduction factors?

- ❑ Capacitance (C)
- ❑ Supply Voltage (V_{DD})
- ❑ Clock frequency (F_{clk})
- ❑ Activity factor (α)

$$P_{SW} = \alpha * C_L * V_{DD}^2 * f_{clk}$$

❑ How you can reduce the activity factor?

- ❑ The proper method is tuning off the clock which drives the un-used register.

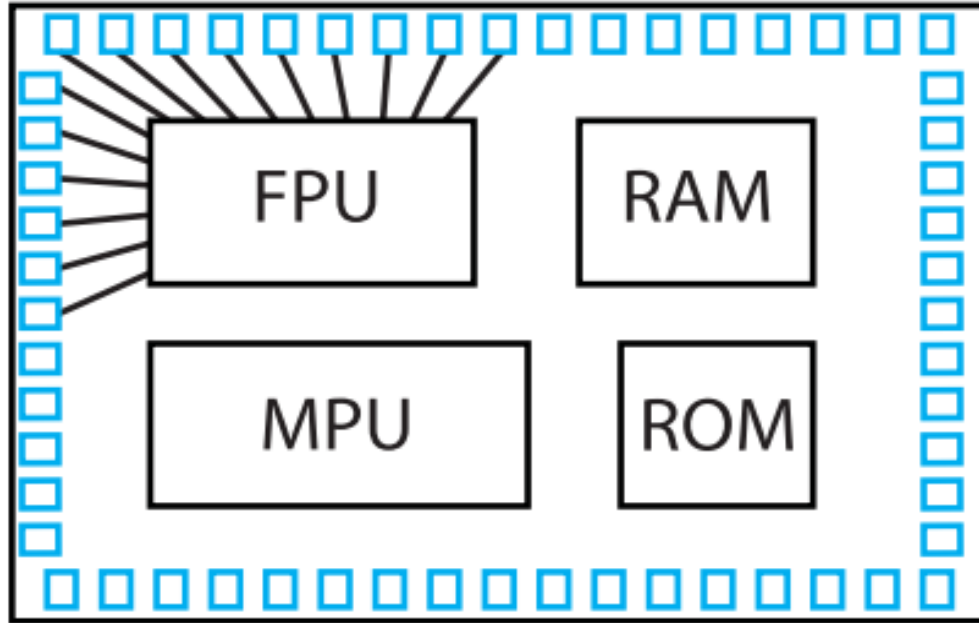


Discussion (2/4)

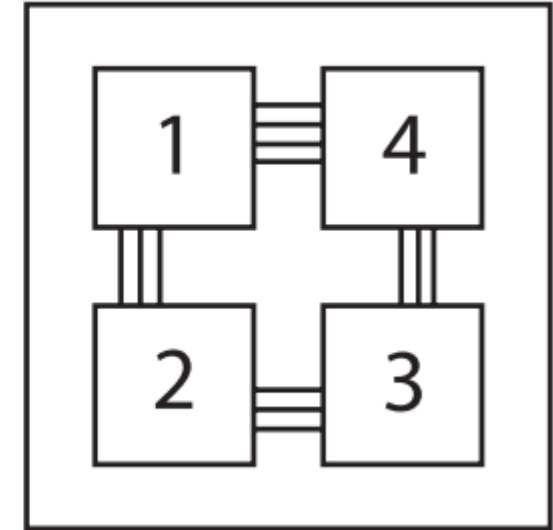
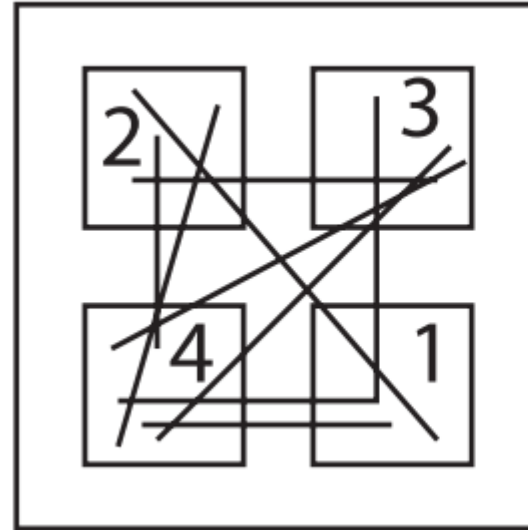


- ☐ **How you can reduce the capacitance of the circuit?**
- ☐ **There are two existing methods to reduce the circuit's capacitances**
 - ☐ **Gate's capacitance**
 - ☐ Use fewer stages of logic gate
 - ☐ Use small size of the gate (W, L)
 - ☐ **Wire capacitance**
 - ☐ Good floor planning is placement of the electronic elements inside the integrated circuit to keep communication blocks close to each other. Thus, the resulting parasitic capacitance will be reduced strongly.
 - ☐ If we had to use long wires, we should use inverters or buffers for driving.

Discussion (3/4)

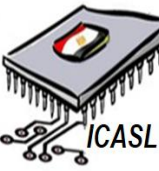


Inputs and outputs can be located near their appropriate cell block

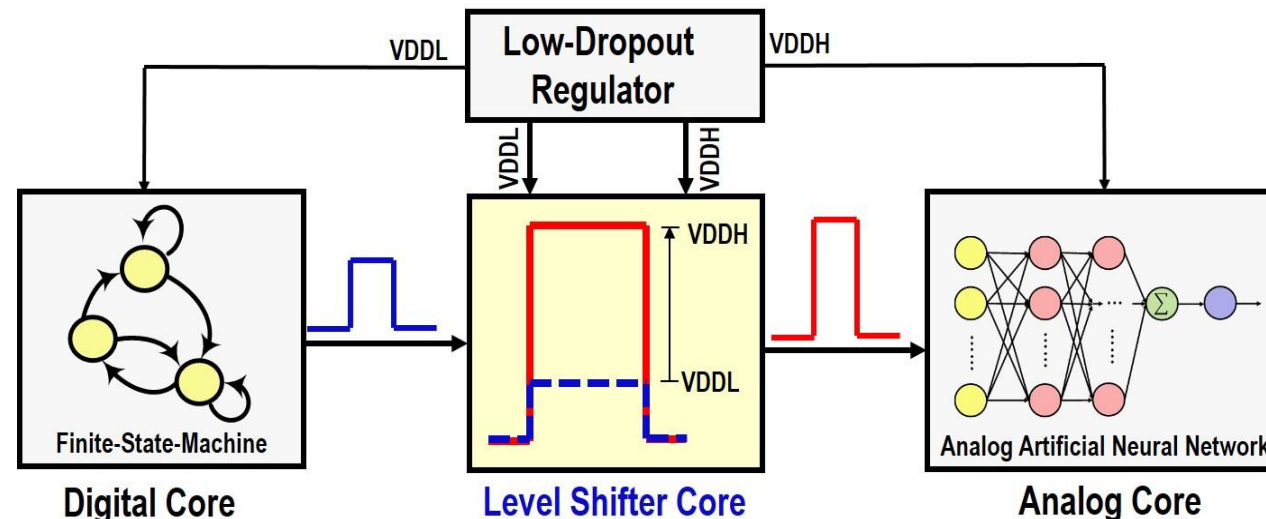


Neat flylines indicate good floor-planning

Discussion (4/4)



- ❑ How you organize the voltage domain (Supply voltage) to reduce the power dissipation?
 - ❑ You can provide separated supply voltage to different blocks
 - ❑ Sometimes, a low power supply can be used for **digital** core while the **analog** core use higher supply voltage.
 - ❑ Usually, a **level shifter** with up/down conversion voltage can be used.



Conceptual Mixed-Mode System-On-Chip

Thanks!